

$$(2)/(1): N' = 0.25 N_v$$

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According to 1st law of thermodynamics, the internal energy of a system depends on the heat absorbed and the work done on the system. Thus, the internal energy of the system can also be changed without heating the system.

For option B and C, it is only true if the system is an ideal gas.

For option D, internal energy also depends on mass of the system. Two systems with same temperature but different masses will have different internal energies.

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